

ROTIMI DAYO

Task 1:

	Rise time (s)	Overshoot	Settling time to 2% (s)	Settling time to 5%
Sports Mode	0.18718	49.2162%	1.9752	1.4086
Cruise Mode	0.23335	56.1712%	3.1391	2.3509

The simulation results for the quarter car suspension system in both sports and cruise modes reveals significant differences in dynamic responses. These differences are influenced by variations in the physical parameters such as spring constants and damping coefficients.

Response Characteristics

The simulation results highlighted this differences between the sports and cruise modes:

1. **Rise Time:**

- **Sports Mode:** 0.18718 seconds
- **Cruise Mode:** 0.23335 seconds
- **Analysis:** The sports mode exhibited a faster rise time, indicating a quicker response to the input disturbance. This is attributed to the stiffer spring constant (K_2), which allows the suspension to react more promptly.

2. **Overshoot:**

- **Sports Mode:** 49.2162%
- **Cruise Mode:** 56.1712%
- **Analysis:** The cruise mode showed a higher overshoot, indicating that the system exceeds the desired displacement more significantly. This is due to the lower damping coefficient (C_2), which results in less resistance to motion and greater susceptibility to overshooting.

3. **Settling Time (2% and 5%):**

- **Sports Mode (2%):** 1.9752 seconds
- **Cruise Mode (2%):** 3.1391 seconds
- **Sports Mode (5%):** 1.4086 seconds
- **Cruise Mode (5%):** 2.3509 seconds

Impact of Parameter Changes: Sports vs. Cruise Mode

Sports Mode:

1. Spring Force (K_2):

- **Higher Spring Constant (K_2):** In sports mode, the higher spring constant results in a greater restoring force for a given displacement, making the suspension system more responsive. However, this will lead to a stiffer ride, potentially reducing comfort because the suspension doesn't absorb bumps as effectively.

2. Damping Force (C_2):

- **Increased Damping Coefficient (C_2):** The increased damping coefficient enhances the resistance to motion, which reduces oscillations and helps the system settle faster. This leads to a more controlled and stable ride, but it might feel harsher because the system resists motion more strongly.

Cruise Mode:

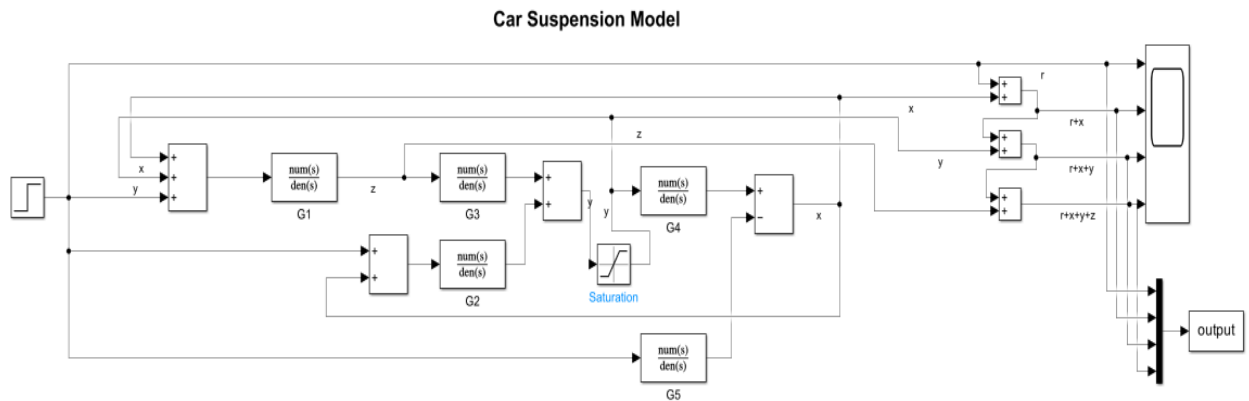
1. Spring Force (K_2):

- **Lower Spring Constant (K_2):** In cruise mode, the lower spring constant provides a softer restoring force for a given displacement. This allows the suspension to absorb bumps more effectively, resulting in a more comfortable ride. However, it may not be as responsive, which can reduce handling precision.

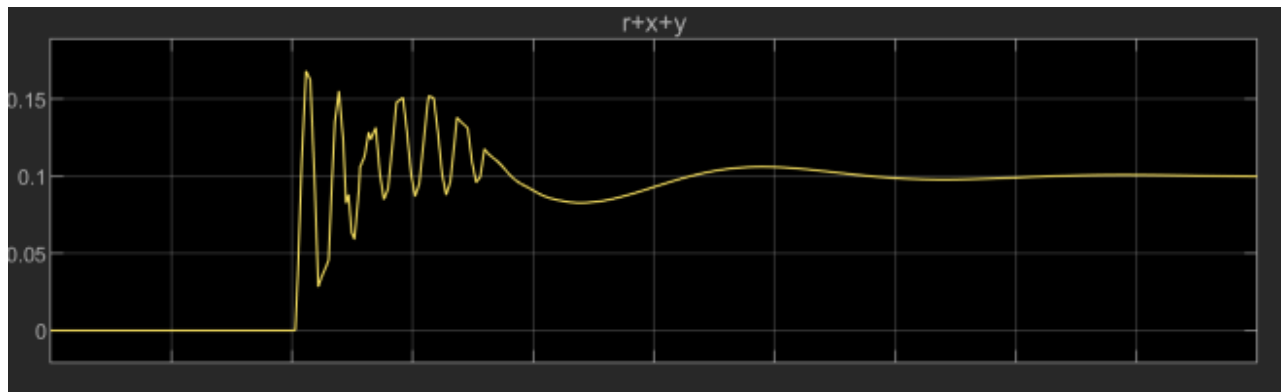
2. Damping Force (C_2):

- **Reduced Damping Coefficient (C_2):** The lower damping coefficient allows for smoother but slower responses. This means the suspension will have higher overshoot and longer settling times, which can make the ride feel more comfortable and less harsh, but may lead to less control during dynamic driving situations.

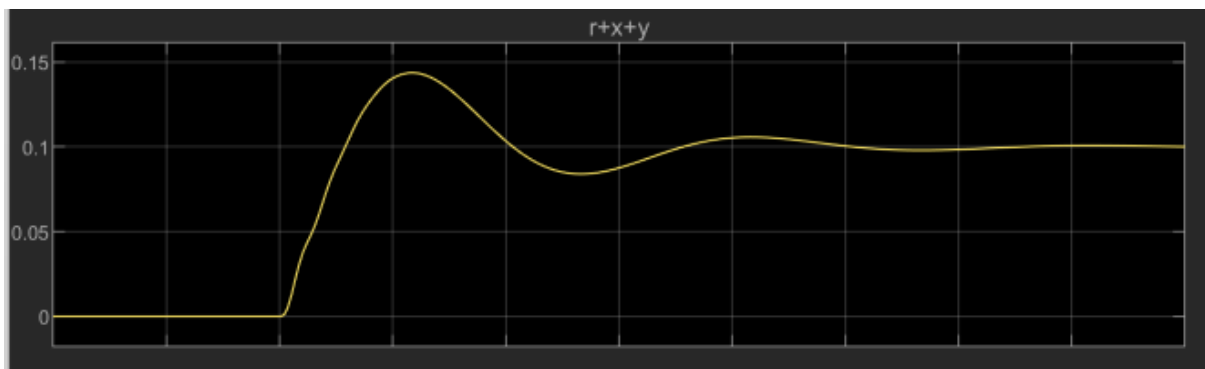
Task 2:



I modified the existing simulink model by adding a **saturation** block for signal **y** because it effectively limits the relative displacement. This means the suspension cannot compress or extend beyond ± 2 cm



With Saturation



Without Saturation

Observations of the Graphs:

- **With Saturation:**
 - The graph shows limited peaks, indicating that the relative displacement is constrained.

- The system response appears more stable with reduced oscillation amplitudes.
- **Without Saturation:**
 - The graph shows larger, smoother peaks, indicating that the relative displacement is not constrained.
 - The system may exhibit larger oscillations, leading to potential instability and discomfort.

Impact on Forces:

- **Spring Force:** The spring force, which depends on the displacement is limited. If the relative displacement is constrained, the maximum spring force is also capped.
- **Damping Force:** Similar to the spring force, the damping force which depends on the rate of change of displacement is also constrained by the saturation limits.

Overall Response of the Vehicle

- **Reduced Amplitude of Oscillations:**
 - By limiting the relative displacement, the amplitude of the oscillations in the suspension is reduced.
 - This results in a more controlled and damped response, preventing excessive motion that could lead to instability.
- **Improved Stability:**
 - Limiting the relative displacement helps in maintaining better control over the vehicle's movement.
 - By preventing excessive suspension travel, the vehicle remains more stable, especially when encountering sudden road irregularities.
- **Force Distribution:**
 - The forces acting on the suspension components are distributed more evenly. The saturation block ensures that the forces do not exceed the structural limits of the suspension system.
 - This can prevent potential damage to the suspension components and enhance their lifespan.

Task 3:

Using the given combinations of damping values C_2 and spring stiffness values K_2 . I Updated the Simulink model parameters for each combination and executed the simulation. I also made sure the wheel displacement and road profile signals were logged during simulations and used MATLAB's stepinfo function to compute the percentage overshoot and settling time for each combination.

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Table 2: Percentage overshoot:

C2 (Ns/m) \ K2(N/m)	5000	13000	30000	50000
1000	87	84.909	80.654	75.204
1500	85.883	82.859	77.809	73.208
3000	85.7	79.813	73.993	66.633
6000	75.697	72.938	67.002	59.95

Table 3: Settling time (2% of Steady-State Value)

C2 (Ns/m) \ K2(N/m)	5000	13000	30000	50000
1000	1.1608	1.0266	1.0676	1.3988
1500	0.8577	0.775	0.7395	1.1197
3000	0.3006	0.3719	0.3817	0.5767
6000	0.3794	0.3702	0.1918	0.2824

Analysis:

Impact of Increasing/Decreasing Damping Value C2:

- **Increasing C2:**
 - **Reduces Overshoot:** Higher damping values provide more resistance to oscillations, reducing the magnitude of the overshoot.
 - **Shortens Settling Time:** Higher damping helps dissipate energy more quickly, allowing the system to stabilize faster.
- **Decreasing C2:**
 - **Increases Overshoot:** Lower damping results in less resistance to oscillations, leading to higher overshoot.
 - **Lengthens Settling Time:** Lower damping means less energy dissipation, causing the system to take longer to settle.

Impact of Increasing/Decreasing Spring Stiffness K2:

- **Increasing K2:**
 - **Increases Natural Frequency:** Higher spring stiffness increases the system's natural frequency, leading to quicker responses.
 - **Potentially Increases Overshoot:** If not adequately damped, the stiffer springs can cause higher overshoot due to more aggressive response.

- **Decreasing K2:**
 - **Decreases Natural Frequency:** Lower spring stiffness reduces the system's natural frequency, leading to slower responses.
 - **Reduces Overshoot:** Less aggressive response reduces the likelihood of high overshoot, provided damping is sufficient.

Selection of Optimal Parameters

We are aiming for a percentage overshoot less than **72%** and a settling time of less than **0.25s**

Optimal Combination:

Based on the analysis, the combination $C2 = 6000 \text{ Ns/m}$ and $K2 = 30000 \text{ N/m}$ satisfies the design constraints:

- **Overshoot:** 67.022%
- **Settling Time:** 0.1918 seconds

This combination ensures minimal overshoot and quick settling.

TASK 4:

Modifying the code by:

- **Transforming Road Profile Data to Timeseries:**
 - I converted the road profile data (`roadprofileL` and `roadprofileR`) into timeseries objects (`roadprofileL_ts` and `roadprofileR_ts`).

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```
roadprofileL_ts = timeseries(roadprofileL, sampledTime);  
roadprofileR_ts = timeseries(roadprofileR, sampledTime);
```

- **Creating a New Simulink Model:**
 - I duplicated the existing Simulink model and replaced the `Step` blocks with `From Workspace` blocks to input the timeseries data directly into the simulation. This ensured the simulation used realistic and time-varying road profiles instead of simple step inputs.
- **Setting Simulation Parameters:**
 - I configured the sampling time of the Simulink model to 1 second to match the sampling rate of the road profile data.

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```
set_param('Whole_Axle_Suspension_Model', 'FixedStep', '1');
```

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Running the Simulation:

- I used the updated Simulink model to run simulations for both sports and cruise modes.

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```
simin =  
Simulink.SimulationInput('Whole_Axle_Suspension_Model');  
simin = simin.setVariable('C2', C2);  
simin = simin.setVariable('K2', K2);  
simin = simin.setVariable('roadprofileL_ts', roadprofileL_ts);  
simin = simin.setVariable('roadprofileR_ts', roadprofileR_ts);  
out = sim(simin);
```

Calculating RMSE and Traction Analysis

- **RMSE Calculation:**

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```
RMSE_left = sqrt(mean((left_wheel_pos -  
roadprofileL_ts.Data).^2));  
RMSE_right = sqrt(mean((right_wheel_pos -  
roadprofileR_ts.Data).^2));
```

- **Traction Analysis:**

- **Cruise Mode:** The lower RMSE values indicate better traction because the suspension allows the wheels to follow the road surface more closely.
- **Sports Mode:** The higher RMSE values indicate reduced traction due to the stiffer suspension settings, which prioritize handling and responsiveness over comfort.

Calculating MAE and Horizontal Stability

- **MAE Calculation:**

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```
MAE_horizontal_stability = mean(abs(left_wheel_pos -  
right_wheel_pos));
```

- **Horizontal Stability Analysis:**
 - **Cruise Mode:** Lower MAE values indicate better horizontal stability because the softer suspension settings allow for synchronized movement of the wheels.
 - **Sports Mode:** Higher MAE values indicate reduced horizontal stability due to the stiffer suspension, which can cause more independent movement of the wheels.

Summary

- **Traction:**
 - **Cruise Mode:** Higher traction with lower RMSE values.
 - **Sports Mode:** Lower traction with higher RMSE values.
- **Horizontal Stability:**
 - **Cruise Mode:** Better stability with lower MAE values.
 - **Sports Mode:** Reduced stability with higher MAE values.